📘 IMS Portal – Institute Management System

A multi-tenant, secure, scalable, AI-ready Institute Management System (IMS) built entirely using op source technologies.

🚀 Project Vision

IMS Portal is a SaaS-ready web platform designed for tuition centers and small institutes to manage:

- Student data
- Study materials
- Assessments (AI & manual)
- Payments tracking
- Notifications
- Attendance (Phase 2)
- Role-based access control

The system is designed with:

- 🔐 High Security
- 📈 Scalability
- 🧠 AI Integration Capability
- 🏢 Multi-Tenant Architecture
- 🎨 Calm, User-Friendly UI
- 🆓 Open-Source Technologies

🏗️ System Architecture Overview

Frontend (Next.js) ↓ Backend (NestJS) ↓ PostgreSQL Database ↓ Redis (Caching & Session Con optional) ↓ File Storage (Local / MinIO) ↓ AI Engine (OpenAI / Ollama - optional)

🧱 Technology Stack

🌐 Frontend

✅ Next.js (React Framework)

Why: - Server-side rendering support - SEO-friendly landing page - Production-ready - Large ecosy
TypeScript support - Ideal for SaaS applications

Why NOT plain React? - No built-in SSR - More manual configuration - Less structured for SaaS

Why NOT Angular? - Heavy for this scale - Steeper learning curve

Why NOT Vue? - Smaller ecosystem compared to React

🎨 UI & CSS

✅ Tailwind CSS

Why: - Utility-first - Lightweight - Highly customizable - Modern SaaS look - Easy theming

Optional: - Shadcn UI (Component library)

Why NOT Bootstrap? - Older UI style - Heavy CSS - Limited customization

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🧠 Backend

✅ NestJS (Node.js Framework)

Why: - Structured architecture - TypeScript-first - Modular design - Scalable - Enterprise-ready

Alternative: - Express.js (simpler but less structured)

Why NOT Django? - Mixing Python backend with JS frontend increases complexity

Why NOT PHP/Laravel? - Less modern SaaS ecosystem - Not ideal for AI integrations

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🗄️ Database

✅ PostgreSQL

Why: - Strong relational support - ACID compliance - Ideal for multi-tenant systems - Reliable fo
& audit logs

Why NOT MongoDB? - IMS requires strong relational data integrity

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🔐 Authentication & Security

- JWT (Access + Refresh Tokens)
- bcrypt (Password hashing)
- Role-Based Access Control
- Feature-Based Access Control
- Single Active Session Enforcement
- Rate Limiting
- Input Validation (Zod / class-validator)
- Helmet (Security headers)
- Soft Delete
- Audit Logs

📁 File Storage

Development: - Local file storage

Production (Open Source): - MinIO (S 3 -compatible)

🤖 AI Integration (Optional in V 1)

Option 1 : - OpenAI API (initial testing)

Option 2 : - Ollama (Self-hosted LLM like LLaMA, Mistral)

🏢 Core Features (Version 1)

🏠 Landing Page

- Portal introduction

- Sign In
- Sign Up

🔐 Authentication

- Admin registration with feature selection
- Student login
- Single active session enforcement

👩‍💼 Admin Features

Dashboard

- Total students
- Pending fees
- Upcoming assessments
- Notifications

Students Management

- Add student
- Edit student
- Soft delete student
- Bulk upload (Excel validation)
- Search & sorting

Study Materials

- CRUD operations
- Hide materials
- Card-based UI
- View-only for students

Assessments

- Manual upload
- AI-generated question papers
- Dynamic validation
- Marking system (MCQ auto, descriptive manual)

Payments

- Track monthly status
- Mark paid/unpaid
- View history

Notifications

- Send to everyone
- Send to specific students
- Unread count

🎓 Student Features

- Login
- Profile management
- Change password
- View materials
- Attempt assessments
- Upload answer sheets
- View marks
- Receive/send notifications

🧠 Advanced Architecture Features

🔹 Multi-Tenant Architecture

- Each institute has unique institute_id
- All queries filtered by institute_id

🔹 Soft Delete

- is_deleted flag
- deleted_at timestamp

🔹 Audit Logs

Tracks: - Student updates - Payment updates - Marks changes - Assessment creation

🔹 Role Hierarchy (Future)

- Super Admin
- Institute Admin
- Teacher
- Student

🔹 Attendance Module (Phase 2)

- Daily attendance marking
- Attendance history

🔐 Load & Security Control

- Rate limiting for login & APIs
- Single active session per user
- Token validation with session_id
- Pagination for large datasets
- Database indexing
- Optional Redis caching

🎨 UI Theme

Design Principles: - Calm pastel color palette - Card-based layout - Minimal sidebar navigation typography (Inter / Poppins) - Soft shadows - Light background (#F5A7) - Primary color (#4F8EF7)

📂 Suggested Project Structure

Frontend

/src /components /pages /layouts /services /styles

Backend

/src /modules /auth /students /assessments /payments /materials /notifications /common /config

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📈 Scalability Plan

Phase 1: - Single backend instance - PostgreSQL

Phase 2: - Add Redis - Add caching

Phase 3: - Add NGINX load balancing - Horizontal scaling

🧪 Development Roadmap

Phase 1 : - Authentication - Multi-tenant base - Students CRUD

Phase 2 : - Study materials - Assessments basic

Phase 3 : - AI integration - Payments

Phase 4 : - Notifications - Attendance

🏁 Conclusion

IMS Portal is designed as:

- A secure multi-tenant SaaS platform
- Fully open-source based
- Scalable and production-ready
- AI-integrated and future-proof

This architecture ensures:

✔ Security ✔ Scalability ✔ Maintainability ✔ Extensibility ✔ Business readiness

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📜 License

Open-source project for educational and startup development purposes.

Built with ❤️ using modern open-source technologies.